

Clinical Natural Language Technology for Health Care

Past, Present, and Future Approaches: NLP, OCR, Computer Vision, LLM, and LMM

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Defining Clinical Natural Language Technology

Clinical Natural Language Technology (clinical NLT) refers to computational methods that allow machines to read, interpret, and act on human language in healthcare settings. The field spans several technique generations: rule-based **Natural Language Processing (NLP)** that matched physician notes against medical vocabularies; **Optical Character Recognition (OCR)** that converted paper records into machine-readable text; **Computer Vision (CV)** applied to radiology images and scanned forms; and, most recently, **Large Language Models (LLMs)** and **Large Multimodal Models (LMMs)** that reason across structured and unstructured data simultaneously. Healthcare imposes uniquely demanding requirements on these systems: clinical language is dense with abbreviations, negations, and coding standards (ICD-10, SNOMED CT, CPT), where a single misread negation in a physician note carries direct financial and compliance consequences.

Relevant Trends

From Rule-Based Systems to Foundation Models

Clinical NLT has evolved through three generations. First-generation tools (1990s to 2010s) such as cTAKES and MedLEE used curated lexicons. Second-generation models like BioBERT and ClinicalBERT introduced transformer architectures and biomedical pre-training. The current generation centers on instruction-tuned LLMs such as Med-PaLM 2 and BioMistral, capable of few-shot extraction and cross-modal reasoning without task-specific retraining. This shift is reshaping the market: NLP in Healthcare and Life Sciences is projected to grow from \$5.18 billion in 2025 to \$16.01 billion by 2030 at a 25.3% CAGR, stimulated by ambient clinical intelligence and AI-powered documentation automation (MarketsandMarkets, 2025).

NLP in Payment Integrity and Risk Adjustment

Payers are deploying NLP directly within claims review and risk adjustment workflows. AI-powered tools now cross-reference encounter data against Hierarchical Condition Category (HCC) codes at submission, identifying coding gaps prospectively rather than through retrospective audit. Cotiviti's Post-Visit Review solution operationalizes exactly this capability, using AI and NLP to reconcile clinical encounter data with HCC codes at scale (Cotiviti, 2025a). NLP is also being used to surface anomalous language patterns in clinical documentation to detect fraud, waste, and abuse (FWA), a high-stakes application demanding both speed and precision.

Opportunities and Threats

Opportunities

- **Unstructured data value:** Roughly 80% of healthcare data is unstructured. LLMs unlock this value for payment accuracy, quality measurement, and population health analytics that rule-based systems cannot reach.
- **Multimodal reasoning:** LMMs jointly process physician notes, lab results, and imaging in one inference call, enabling holistic documentation audits no prior generation of technology could support.
- **Real-time coding accuracy:** NLP-assisted HCC coding and CPT validation can run continuously at claim volume, far exceeding the consistency of manual audit teams and directly improving payer financial performance.

- **Interoperability:** FHIR-native NLP pipelines harmonize data across fragmented EHR systems, a capability Cotiviti significantly expanded through its 2025 acquisition of Edifecs (Cotiviti, 2025b).

Threats

- **Hallucination risk:** LLMs can produce plausible but factually incorrect output. In payment integrity contexts, a fabricated audit finding carries real legal and financial liability, requiring human-in-the-loop review architectures.
- **HIPAA compliance:** Routing Protected Health Information through cloud-hosted LLMs introduces regulatory risk, often requiring on-premise or private-cloud deployments that add cost and complexity.
- **Model bias:** Models trained on academic medical center data may underperform on records from rural or underserved populations, introducing systematic errors in risk scoring and quality measurement.

Strategic Recommendations for Cotiviti

Cotiviti's mission to enable a high-quality and viable healthcare system, combined with its depth in payment integrity data and the interoperability infrastructure from the Edifecs acquisition, positions it uniquely to lead the next generation of clinical NLT. Two investments stand out.

1. Clinically Fine-Tuned LLM for Prospective Documentation Gap Detection. Cotiviti should invest in fine-tuning a domain-specific LLM on its proprietary corpus of encounter data, HCC coding outcomes, and audit decisions. A model trained on this closed-loop signal would identify documentation gaps at claim submission, shifting the value proposition from retrospective audit to prospective accuracy. This extends the logic already validated in Post-Visit Review into a faster, more scalable form that no competitor without Cotiviti's data depth could replicate (Cotiviti, 2025a).

2. Responsible AI Framework as a Market Differentiator. Cotiviti's leadership has stated that AI should amplify expert judgment rather than replace it (Cotiviti, 2025c). Formalizing this as a published Responsible Clinical AI framework, covering model explainability, clinician review checkpoints, bias monitoring, and audit trails, would directly address health plans' growing liability concerns and reinforce Cotiviti's differentiated values of transparency, authenticity, and accountability (Cotiviti, 2025d). In a market crowded with AI vendors making broad promises, this framework would make Cotiviti the trusted choice.

Why Cotiviti, and What I Can Contribute

What drew me to Cotiviti is that it operates where data quality directly affects both patient outcomes and system financial integrity. Cotiviti's approach to responsible innovation especially stands out to me: AI as infrastructure that makes expert clinical judgment more effective and scalable, not a substitute for it. Seeing that philosophy reflected in Cotiviti's leadership at CCC25 and in products like Post-Visit Review confirmed that this is a company working on problems that genuinely matter. As an ECE Honors and Mathematics student with a Finance minor at UT Austin, I sit at an intersection that maps directly onto Cotiviti's work. My engineering and mathematics background, particularly in probability and statistical inference, helps me reason clearly about model reliability, false positive rates, and the real-world consequences of error in a claims context. My finance training adds a business lens: the ability to frame AI investments in terms of ROI, risk exposure, and competitive positioning. My Clinical Report AI Assistant, the Proof of Concept I built for this assessment, reflects that approach by summarizing clinical notes, extracting conditions, medications, and tests, flagging coding and documentation gaps, and supporting interactive Q&A. I am specifically excited to contribute to NLP pipelines, risk adjustment tools, or AI governance work at Cotiviti, areas where I believe the company's most consequential near-term opportunities lie.

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